Word Level Autoregressive Limerick Generation System

Foundations of Generative AI ICT4403 Assignment 2

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System title	Word Level Limerick Generation with RNN LSTM and GRU
Dataset	Limerick Dataset OEDILF v3 - https://doi.org/10.5281/zenodo.5722527
GitHub repository	https://github.com/mizar-0/RNN-LSTM-GRU-Comparison

Executive summary

Four word-level autoregressive models were trained on the filtered OEDILF limerick corpus under the same data split and core hyperparameters. GRU produced the lowest held-out perplexity at 45.30 and the highest top-1 and top-5 accuracy. The final generation setting is GRU at temperature 0.5: its strict AABBA rhyme score was 33.67%, close to 34.25% at temperature 0.3, while repeated trigrams fell from 3.85% to 0.78% and distinct-2 rose from 0.241 to 0.415. Generated poems often contained fluent fragments but rarely formed a coherent setup and punchline. The blinded qualitative scores were produced by Claude as an LLM judge, not by a human rater.

1 Problem and objective

The task was to build an autoregressive text-generation system using Simple RNN, Bidirectional RNN, LSTM, and GRU architectures. Each model predicts the next word from its preceding context and generates a poem one token at a time from a seed sequence. The comparison covers predictive performance, computational cost, generated-text quality, and the effect of sampling temperature.

2 Dataset and preprocessing

The Limerick Dataset OEDILF v3 was obtained from Zenodo under the CC BY 3.0 US licence. The source contains 98,454 records. Filtering by the supplied limerick label retained 65,886 five-line poems from 1,296 authors. Poems contain a mean of 29.76 words and a median of 30; the 10th and 90th percentiles are 26 and 34 words. The material is user-submitted and includes some crude, offensive, or adult language.

The poems were shuffled with seed 42 and split by poem into 80% training, 10% validation, and 10% test partitions before sequence construction. Text was lowercased and tokenised into words and punctuation. Each poem received beginning-of-sequence, end-of-line, and end-of-sequence tokens. The 10,000-token training vocabulary retained pad and unknown tokens. About 10.16% of lexical training tokens and 9.92% of lexical test tokens map to the unknown token. The frequency distribution is concentrated in function words; the five most frequent lexical tokens are the, a, and, to, and in.

A 64-token left context was paired with each target token, yielding 2,229,796 training, 279,120 validation, and 278,760 test examples. Splitting before window construction prevents token windows from the same poem appearing in more than one partition.

3 Model architectures

All four models shared the same input representation and output layer. A 128-dimensional embedding maps token indices to dense vectors. The recurrent layer uses 256 hidden units and dropout of 0.3. A linear projection then maps the recurrent representation to the 10,000-token vocabulary.

Model	Recurrent layer	Output width	Sequence representation	Parameters
Simple RNN	RNN 256	256	Final forward hidden state	3.95M
BiRNN	Bidirectional RNN 256	512	Concatenated forward and backward states	6.61M
LSTM	LSTM 256	256	Final hidden state	4.25M
GRU	GRU 256	256	Final hidden state	4.15M

4 Experimental protocol

The models used Adam with a learning rate of 0.001, batch size 512, context length 64, and random seed 42. Training was capped at 40 epochs. Validation-loss early stopping used a patience of five epochs, and the lowest-validation-loss checkpoint was retained. The final checkpoint for each model was evaluated once on the held-out test set. Because early stopping is part of the protocol, models trained for different numbers of epochs; the comparison uses each model's best validated checkpoint rather than its final epoch.

5 Training behaviour

The learning curves show that LSTM and GRU reached lower validation loss than the two vanilla recurrent models. Simple RNN stopped after 22 epochs with its best checkpoint at epoch 17. BiRNN stopped after 33 epochs with its best checkpoint at epoch 28. LSTM stopped after 26 epochs with its best checkpoint at epoch 21. GRU reached the 40-epoch cap and its best checkpoint was the last epoch, indicating that it had not triggered early stopping.

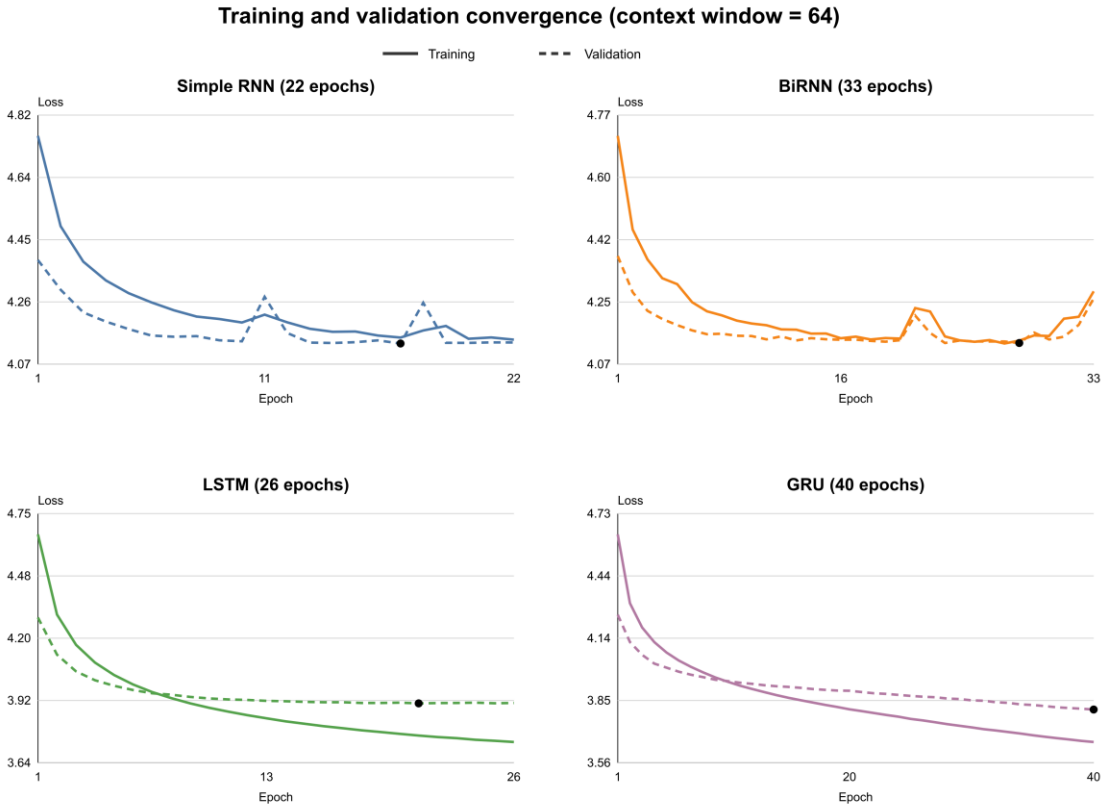


Figure 1 Training and validation loss for the final context length 64 runs

5 1 Architecture based interpretation

The observed ranking is consistent with the models' memory mechanisms, but it does not prove that architecture alone caused every difference. The experiment used one seed, validation-based stopping produced unequal epoch counts, and GRU was still improving at epoch 40. The following explanations should therefore be read as mechanisms that fit the evidence rather than isolated causal effects.

Simple RNN repeatedly compresses the 64-token history into one tanh hidden state. Older information and its gradients can decay over many recurrent steps, so the model can learn frequent local word transitions while struggling to retain cues across several limerick lines. Its relatively high perplexity and near-chance rhyme score are consistent with that limitation.

BiRNN reads the same completed context window in both directions and concatenates two 256-unit states. It never sees the target word or future generated text, so the backward pass supplies another ordering of the existing past rather than the future context that usually makes bidirectional models useful. The concatenated state also doubles the input to the 10,000-word output projection, raising the parameter count and per-epoch cost. This explains why it was much more expensive than Simple RNN without gaining perplexity; the reverse state may also give less direct emphasis to the most recent tokens needed for autoregressive prediction and line termination.

LSTM uses a cell state with input, forget, and output gates, providing a more stable route for information and gradients than a vanilla RNN. This mechanism is consistent with its lower perplexity and stronger blinded coherence scores. It does not, however, explicitly plan an AABBA rhyme scheme. Average next-word loss is dominated by common tokens, while rhyming line endings are sparse decisions separated by many steps, so better token prediction need not produce strong strict rhyme.

GRU uses update and reset gates to retain useful history while adapting its candidate state, with no separate cell state and fewer parameters than this LSTM. Its simpler gated path may have been easier to optimise on this dataset and may have retained line-position and earlier-ending cues strongly enough to reproduce rhyme patterns. At low temperature, sampling repeatedly selects the most probable version of those learned patterns, which is consistent with GRU combining high rhyme with the highest repeated-trigram rate and repeated endings such as man at temperature 0.3. Raising the temperature to 0.5 reduces that over-concentration while preserving most of the rhyme score.

6 Held out test results

Model	Epochs	Loss	PPL	Top 1	Top 5	Sec epoch	Total time
Unigram	-	5.702	299.59	11.46%	31.60%	-	-
Simple RNN	22	4.140	62.79	31.94%	50.56%	59.5	21.8 min
BiRNN	33	4.140	62.81	32.00%	50.61%	93.9	51.7 min
LSTM	26	3.913	50.03	34.14%	53.55%	68.9	29.9 min
GRU	40	3.813	45.30	34.84%	54.79%	70.5	47.0 min

The add-one-smoothed unigram baseline confirms that all recurrent models learned useful sequence information. GRU achieved the strongest predictive result: its perplexity of 45.30 was 9.5% lower than LSTM and about 27.9% lower than Simple RNN. It also obtained the highest top-1 accuracy at 34.84% and top-5 accuracy at 54.79%. LSTM and GRU required similar compute per epoch at 68.9 and 70.5 seconds. LSTM's 17.1-minute lower total reflects earlier validation-based stopping, not materially faster epochs. BiRNN used the most compute per epoch without improving perplexity over Simple RNN.

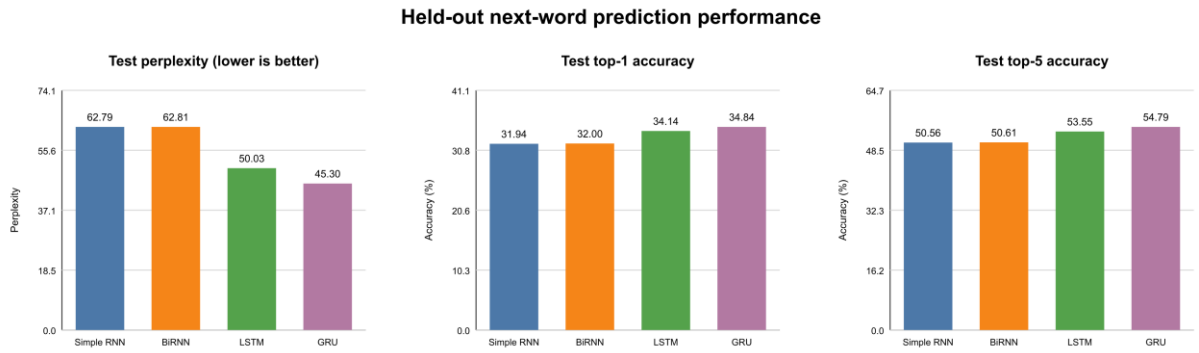


Figure 2 Held-out perplexity and next-word accuracy

7 Generation sweep

The generation sweep produced 7,200 poems from four models, six temperatures, and 300 poems per model-temperature cell. All models used the same five prompts. Automatic metrics covered termination, exact five-line form, pronunciation-dictionary coverage, AABBA rhyme, repeated trigrams, immediate repetitions, copied training lines, distinct n-grams, and unique-poem share. Metric uncertainty was estimated with 2,000 bootstrap resamples.

Strict AABBA rhyme is the mean of four required pair checks: lines 1-2, 3-4, 1-5, and 2-5. A pair scores one only when both end words are distinct, both are covered by the CMU Pronouncing Dictionary, and the dictionary lists them as rhymes. Identical endings, uncovered pairs, and missing five-line structure score zero. Rhyme coverage is the share of the four required pairs containing distinct, dictionary-covered words.

Distinct-2 is the number of unique lexical bigrams divided by the total number of lexical bigrams in a model-temperature cell. The calculation removes control tokens, including beginning-of-sequence, end-of-line, end-of-sequence, padding, and unknown tokens, as well as punctuation. Repeated-trigram rate is calculated per poem as the number of trigram occurrences beyond the first occurrence of each lexical trigram, divided by the number of lexical trigrams; the table reports the mean across 300 poems. These metrics therefore measure lexical sequences and do not include the end-of-line token.

Under this definition, GRU scored 34.25% at temperature 0.3 and 33.67% at 0.5. The held-out human reference scored 94.25% with 94.33% eligible-pair coverage.

8 Temperature effects

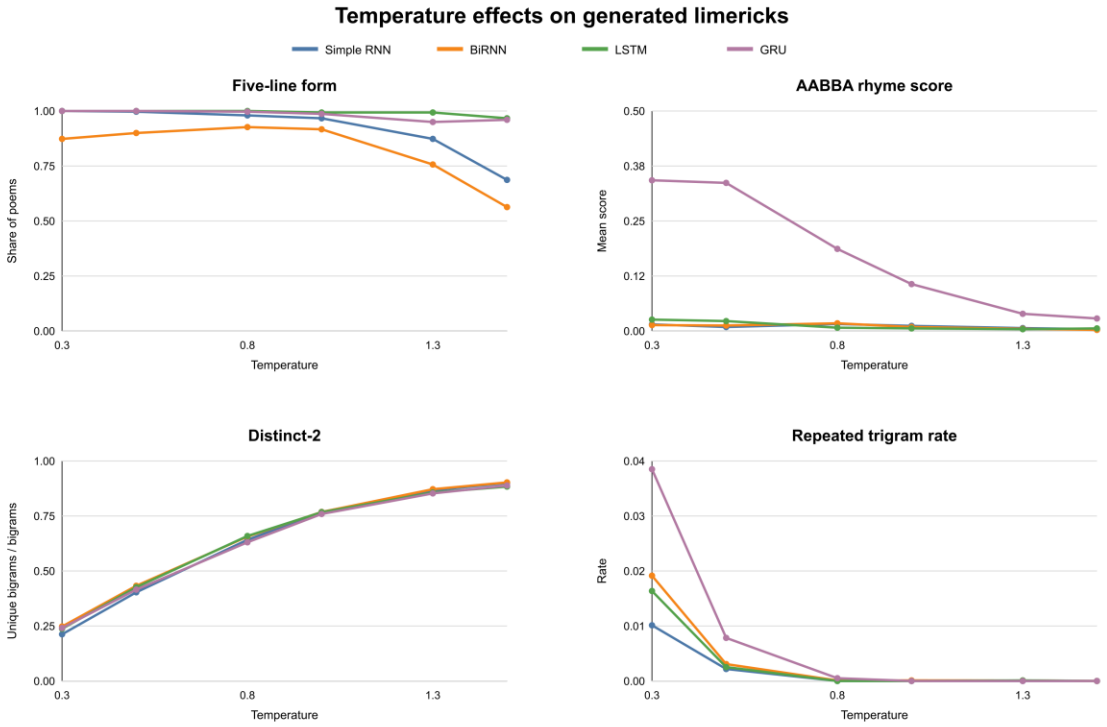


Figure 3 Form quality strict rhyme diversity and repetition across temperatures.

Temperature controlled a structure-diversity trade-off. Across the four models, distinct-2 rose from approximately 0.21-0.25 at temperature 0.3 to 0.88-0.90 at 1.5, while rhyme and five-line form generally weakened. For GRU, moving from 0.3 to 0.5 reduced strict rhyme by only 0.58 percentage points, from 34.25% to 33.67%, but repeated trigrams fell from 3.85% to 0.78% and distinct-2 rose from 0.241 to 0.415. The word man appeared as a line ending 85 times among the 1,500 GRU line endings at temperature 0.3, compared with 39 at 0.5. These results favour 0.5 as the better balance.

Model	T	Five-line	Strict rhyme	Distinct-2	Repeated trigram
Simple RNN	0.3	100.0%	1.50%	0.212	1.01%
Simple RNN	0.5	99.7%	0.92%	0.403	0.22%
Simple RNN	1.0	96.7%	1.17%	0.769	0.00%
BiRNN	0.3	87.3%	1.33%	0.248	1.91%
BiRNN	0.5	90.0%	1.25%	0.433	0.31%
BiRNN	1.0	91.7%	0.92%	0.768	0.01%
LSTM	0.3	100.0%	2.58%	0.237	1.64%
LSTM	0.5	100.0%	2.25%	0.426	0.25%
LSTM	1.0	99.3%	0.58%	0.767	0.00%
GRU	0.3	100.0%	34.25%	0.241	3.85%
GRU	0.5	100.0%	33.67%	0.415	0.78%
GRU	1.0	98.7%	10.67%	0.759	0.00%

Table 3 Numeric generation metrics at three temperatures for every model.

9 Blinded manual evaluation

The blinded sheet contained 80 poems: four models, four temperatures, and five poems per cell. Claude served as an LLM judge and scored coherence and contextual consistency from 1 to 5 without model or temperature labels in the scoring sheet. Table 4 reports the two criteria separately at temperatures 0.3 and 0.5; the combined score is their mean. The complete four-temperature summary is included in the repository. No independent human rating or human-LLM agreement check was collected, so these values are supplementary qualitative evidence rather than a human evaluation.

Model	T	Coherence	Contextual consistency	Combined
Simple RNN	0.3	2.6	2.0	2.3
Simple RNN	0.5	2.0	2.0	2.0
BiRNN	0.3	2.4	2.2	2.3
BiRNN	0.5	2.4	2.4	2.4
LSTM	0.3	3.0	3.0	3.0
LSTM	0.5	2.6	2.6	2.6
GRU	0.3	2.6	3.4	3.0
GRU	0.5	2.6	2.8	2.7

Table 4 Mean blinded Claude LLM judge scores with five poems per cell.

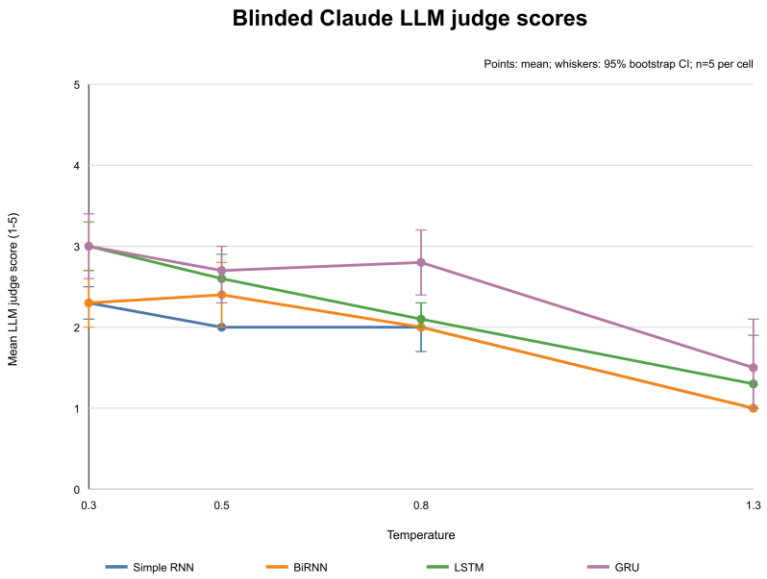


Figure 4 Mean blinded Claude LLM judge scores by model and temperature

10 Generated poem comparison

Table 5 uses the same prompt and sample index for every model at three temperatures. Slashes mark line breaks. The examples show the aggregate pattern: low temperatures preserve recurrent phrasing and form, whereas higher temperatures increase variety but often weaken grammar. Even the strongest outputs are usually fluent fragments rather than complete limericks with a coherent setup and punchline.

Training from scratch on about 66,000 poems provided no pretrained language knowledge. Next-token likelihood encourages local fluency but does not plan rhyme, narrative setup, or punchline. The 10,000-token cap maps about 10% of lexical tokens to the unknown token, removing rare content words; together, these constraints permit recognisable form but make reliable literary quality unlikely.

Model and temperature	Generated poem from prompt there once was a
Simple RNN 0.3	there once was a fellow who knew that he said but she was so i see i am not in a bind on the floor then i fear that i find that it is a great deal
BiRNN 0.3	there once was a woman named fred and a rabbit one made winning bred in the surf and the rest was a victim of fame and a tale of the name of the key
LSTM 0.3	there once was a young lady named who was caught by the long of the day she was born for a week and she said i don't know i don't care but i'm sure i don't know
GRU 0.3	there once was a man of a man and a man who was caught in a pan he was born with a beard and his beard was a beard in a pan when he ran off the pan
Simple RNN 0.5	there once was a girl and a bit well i've got a great deal i don't know i will not have a hunch that i want to be far to my mom your dear friend i would say
BiRNN 0.5	there once was a good to sized with a gem loving type of up but this limerick is banned we've got much more than fun and the rest of this place is the same
LSTM 0.5	there once was a woman named booth who was shot by the moon up named clyde who was built in the past but that meant to be brief he would say as he claimed it i fear
GRU 0.5	there once was a fellow named andy who ate his new car and his mother had a talent to make and his wife made a man and the rest of his daughter was sloppy
Simple RNN 1.0	there once was a top of a thought when the stage is a genius when he was an instrument whoa why that gold is to blame here's a count and its flag look a bit
BiRNN 1.0	there once was a kind out of dough he's a knife and as an affair pace this good end is i need to you keeping you nice these apartment fires so
LSTM 1.0	there once was a lady named grace who exclaimed the emotions you rot if you often can say eating friction a way he will help you fall in it will thrive
GRU 1.0	there once was a number the phrase that produced with the straight cheese of wine it's divine and fresh duty with a chomp and through water the sole part of the more mouth agape

11 Model selection

Model	Prediction	Seconds epoch	Five-line form	Strict rhyme	Decision
Simple RNN	Weak	59.5	99.7% at 0.5	0.9% at 0.5	Fast baseline
BiRNN	Weak	93.9	90.0% at 0.5	1.3% at 0.5	Not selected
LSTM	Strong	68.9	100% at 0.5	2.3% at 0.5	Prediction alternative
GRU	Best	70.5	100% at 0.5	33.7% at 0.5	Selected model

Table 6 Model-selection evidence with every generation metric shown at temperature 0.5.

GRU is selected on two grounds. First, it has the strongest held-out next-word prediction, with the lowest test loss and perplexity and the highest top-1 and top-5 accuracy. Second, the generation sweep provides converging evidence: GRU is the only architecture with substantial strict rhyme at the common temperature of 0.5. After selecting the architecture, its temperature is chosen by balancing rhyme, diversity, and repetition. Temperature 0.5 is preferred to 0.3 because strict rhyme differs by only 0.58 percentage points while diversity improves and repeated trigrams fall sharply. Five-line completion is not a deciding advantage because both GRU and LSTM reached 100% at 0.3 and 0.5.

12 Limitations

The study used one random seed, so training variance is unknown. Validation-based early stopping produced different epoch counts: LSTM stopped after 26 epochs, whereas GRU reached the 40-epoch cap. Their per-epoch times were similar, but the unequal number of updates confounds architectural and total-training comparisons. GRU might improve with a longer cap. The strict rhyme metric depends on CMU Pronouncing Dictionary coverage and treats uncovered or identical word pairs as failures. Qualitative scoring used one Claude LLM judge and five poems per cell, with no independent human agreement check. Although the scoring sheet omitted model and temperature labels, the judge had previously seen some labelled examples from the same generation sweep, so recognition cannot be ruled out and the blinding was incomplete. Perplexity does not directly measure humour, narrative coherence, or originality.

13 Reproducibility

The repository contains preprocessing and training code, run metadata, learning curves, test metrics, the 7,200-sample generation sweep, automatic metric outputs, the blinded scoring workbook, and the unblinded LLM-judge score summary. The raw dataset and model checkpoints are excluded because of their size; download and reproduction instructions are provided. The final configuration records seed 42, data split, vocabulary size, context length, optimiser settings, epoch history, and checkpoint selection for each model.

14 Conclusion

LSTM and GRU outperformed Simple RNN and BiRNN on held-out next-word prediction, but only GRU showed substantial strict rhyme adherence. Because GRU also trained for more epochs than LSTM and only one seed was used, the experiment cannot attribute every difference to architecture alone. GRU at temperature 0.5 is the strongest overall configuration under the stated selection rule. Its outputs remain well below human-reference rhyme quality and generally lack a coherent limerick punchline.

References

Limerick Dataset OEDILF v3. Zenodo. https://doi.org/10.5281/zenodo.5722527

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PyTorch. https://pytorch.org